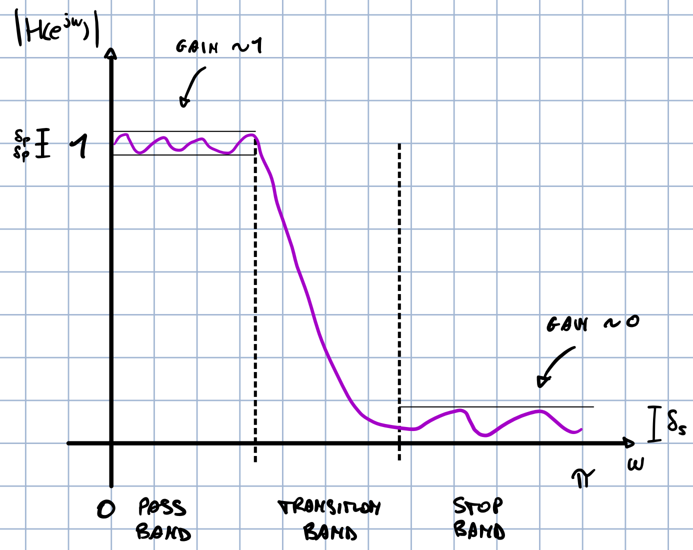
THE DESIGN OF A FREQUENCY SELECTIVE FILTER STARTS WITH A SPECIFICATION OF THE DESIRED FREQUENCY RESPONSE.

A GOOD FILTER SHOULD HAVE:

- SMALL RIPPLE IN PASS-BAND $\quad \omega \in P.B.: \quad 1-\delta_p \le |H(e^{j\omega})| \le 1+\delta_p \qquad 0 \le \delta \ll 1$
- HIGH ATTENUATION IN STOP-BAND $\quad \omega \in S.B.: \quad |H(e^{j\omega})| \le \delta_s$



RELATIVE AMPLITUDES W.R.T. MAX VALUE (NORMALIZED)

PASS BAND	STOP BAND
$\frac{1-\delta_p}{1+\delta_p} \le \frac{\lvert H(e^{j\omega})\rvert}{1+\delta_p} \le 1$	$\frac{\lvert H(e^{j\omega})\rvert}{1+\delta_p} \le \frac{\delta_s}{1+\delta_p}$

NORMALIZED RESPONSES

ATTENUATIONS

$$A_p = 20 \log_{10} \frac{1-\delta_p}{1+\delta_p} \quad [dB]$$

$$A_s = 20 \log_{10} \frac{\delta_s}{1+\delta_p} \quad [dB]$$

FREQUENCY LINK

UP UNTIL NOW WE HAVE BEEN WORKING WITH $\omega \in [-\pi, \pi]$, BUT WE NEED A WAY TO EXPRESS THE FREQUENCY IN Hz. THIS IS DONE THROUGH THE SAMPLING FREQUENCY f_o.

$2\pi : f_c = \omega_x : f_o$

$$\omega_p = 2\pi \frac{f_p}{f_o} \qquad \omega_s = 2\pi \frac{f_s}{f_o}$$

IIR FILTER DESIGN

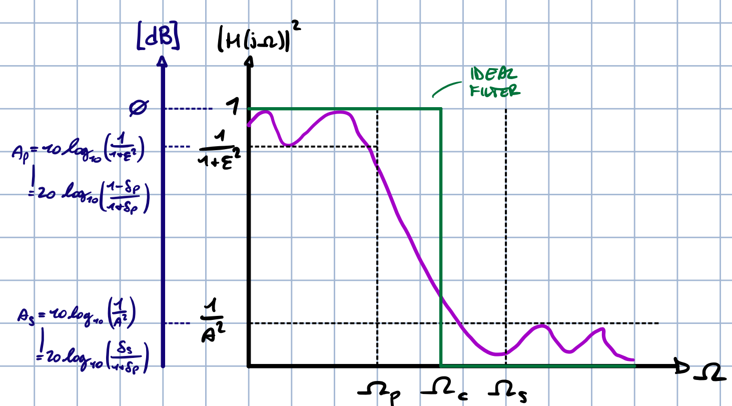
DISCRETE TIME IIR FILTERS HAVE INFINITELY LONG RESPONSES AND CAN BE DESIGNED USING CONTINUOUS TIME FILTER PROTOTYPES.

THIS IS DONE IN THREE STEPS:

1) TRANSFORM THE DISCRETE TIME SPECIFICATION INTO CONTINUOUS
2) DESIGN A CONTINUOUS TIME FILTER
3) CONVERT THE CONTINUOUS TIME FILTER TO DISCRETE

CONTINUOUS TIME DESIGN SPEC.

PREVIOUSLY WE LOOKED AT THE AMPLITUDE RESPONSE W.R.T ω $|H(e^{j\omega})|$, NOW INSTEAD WE USE A DIFFERENT DOMAIN $|H(j\Omega)|^2$, WHERE $j\Omega$ IS THE IMAGINARY PART OF THE LAPLACE S



$$\frac{1}{1+\varepsilon^2} \le |H(j\Omega)|^2 \le 1 \quad , \quad |\Omega| \le \Omega_p$$

$$0 \le |H(j\Omega)|^2 \le \frac{1}{A^2} \quad , \quad |\Omega| \ge \Omega_s$$

PASS-BAND EDGE

STOP-BAND EDGE

$$|H(j\Omega)|^2 = \frac{1}{1+\varepsilon^2} \quad @ \; \Omega = \Omega_p$$

$$|H(j\Omega)|^2 = \frac{1}{A^2} \quad @ \; \Omega = \Omega_s$$

$$A_p = 10\log_{10}\left(\frac{1}{1+\varepsilon^2}\right) \Longrightarrow \varepsilon = \sqrt{10^{-\frac{A_p}{10}} - 1} \longrightarrow \sqrt{\frac{1}{1+\varepsilon^2}} = \frac{1-\delta_p}{1+\delta_p} \Rightarrow \varepsilon = \frac{2\sqrt{\delta_p}}{1-\delta_p}$$

$$A_s = 10\log_{10}\left(\frac{1}{A^2}\right) \Longrightarrow A = \sqrt{10^{-\frac{A_s}{10}}} \longrightarrow \frac{1}{A} = \frac{\delta_s}{1+\delta_p} \Rightarrow A = \frac{1+\delta_p}{\delta_s}$$

LAPLACE VS. Z

$j\Omega$ IS THE IMAGINARY PART OF s

$$H(s)\Big|_{s=\sigma+j\Omega} = \int_{-\infty}^{+\infty} h(t)e^{-st}\,dt$$

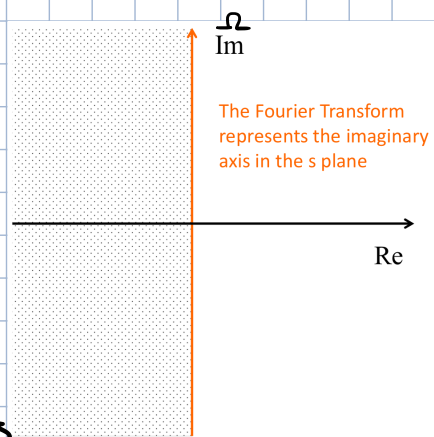
$$H(j\Omega) = H(s)\Big|_{s=\phi+j\Omega} = \int_{-\infty}^{+\infty} h(t)\,e^{-j\Omega t}\,dt$$

ON THE IMAGINARY AXIS

$$h(t) := e^{-\alpha t}u(t) \iff H(s) = \int_{-\infty}^{+\infty} e^{-\alpha t}u(t)\,e^{-st}\,dt = \int_{0}^{\infty} e^{-(\alpha+s)t}\,dt = \frac{1}{s+\alpha}$$

THE LAPLACE TRANSFORM CAN BE EXPRESSED AS A FRACTION OF POLYNOMIALS IN s (IN THIS CASE WITH A POLE IN $s=-\alpha$).

$$H(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k} = \frac{b_0}{a_0}\,\frac{\prod_{k=1}^{M}(s-z_k)}{\prod_{k=1}^{N}(s-p_k)}$$



THE FOURIER TRANSFORM (SAME AS UNIT CIRCLE OF THE Z-TRANSFORM) IS MAPPED TO THE IMAGINARY AXIS Ω

$\longrightarrow$ TO HAVE CAUSAL AND STABLE SYSTEMS WE WANT ALL POLES TO BE ON THE LEFT SIDE

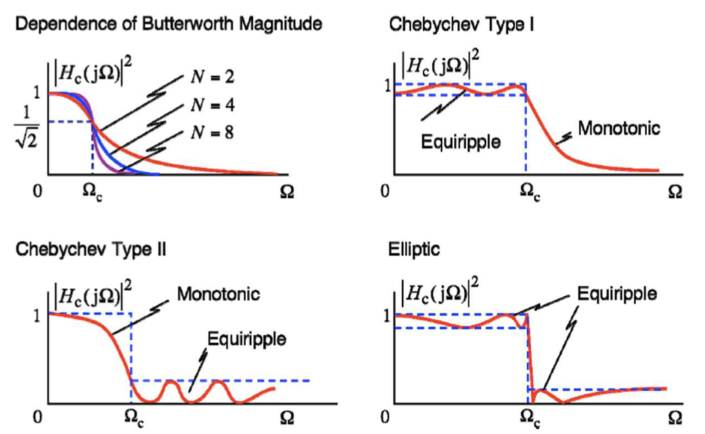
$$\mathrm{Re}\{p_k\} < 0 \quad \forall k$$

THE GOAL IS TO DESIGN THE FILTER IN $|H(j\Omega)|^2$ AND THEN MAP IT TO $|H(e^{j\omega})|$

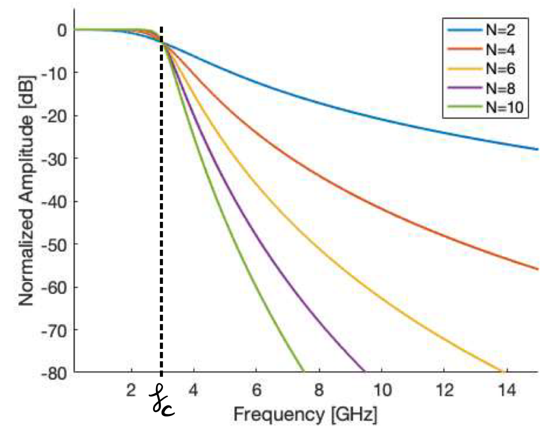
LOW-PASS FILTERS

THERE ARE KNOWN FORMULAS TO DESIGN LPFs

Name	Pass-band	Stop-band
Butterworth	Flat	Flat
Cebyshev I	Ripples	Flat
Cebyshev II	Flat	Ripples
Elliptical	Ripples	Ripples



BUTTERWORTH



$$|H(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

N = ORDER

INCREASING N (ORDER) MAKES TRANSITION SHARPER.
$N \to \infty$ APPROX. IDEAL FILTER (MAGNITUDE)

$$|H(j\Omega)|^2\Big|_{\Omega\to 0} = 1$$

$$|H(j\Omega)|^2\Big|_{\Omega=\Omega_c} = \frac{1}{1+(1)^{2N}} = \frac{1}{2} \to -3\text{dB}$$

! NO RIPPLES IN BOTH PASS- AND STOP-BANDS

$$|H(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}} \longrightarrow |H(s)|^2 = \frac{1}{1+\left(\frac{s}{j\Omega_c}\right)^{2N}}$$

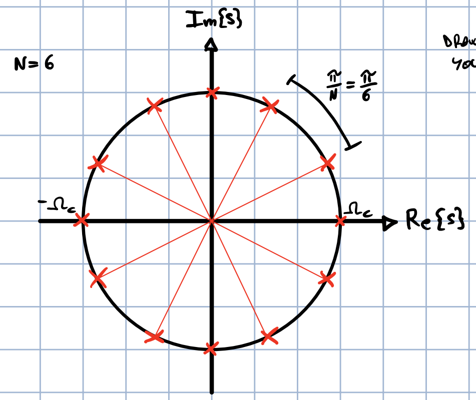
TO STUDY POLES:

$$1+\left(\frac{s}{j\Omega_c}\right)^{2N} = \varnothing \longrightarrow \left(\frac{s}{j\Omega_c}\right)^{2N} = -1 = e^{\overbrace{-j\pi}^{-1}+\overbrace{j2k\pi}^{\text{PERIODIC}}} = e^{j(-\pi+2k\pi)} \quad , \; k=1,\dots,2N$$

$$\left[\left(\frac{s}{j\Omega_c}\right)^{2N}\right]^{\frac{1}{2N}} = \left[e^{j(-\pi+2k\pi)}\right]^{\frac{1}{2N}} \longrightarrow \frac{s}{j\Omega_c} = e^{\frac{j(-\pi+2k\pi)}{2N}} \longrightarrow s = j\Omega_c e^{\frac{j(-\pi+2k\pi)}{2N}}$$

$$j = e^{j\frac{\pi}{2}} = e^{j\frac{\pi}{2}\frac{2N}{2N}}$$

$$S = \Omega_c\, e^{j\frac{\pi}{2}\frac{2N}{2N}}\, e^{\frac{j(-\pi + 2k\pi)}{2N}} = \Omega_c\, e^{j\frac{\pi}{2N}(2k-1)}\, e^{j\frac{\pi}{2N}N} = \Omega_c\, e^{j\frac{\pi}{2N}(2k-1+N)}, \quad k = 1, \dots, 2N$$



DRAWING IS FUCKED BUT YOU GET THE POINT

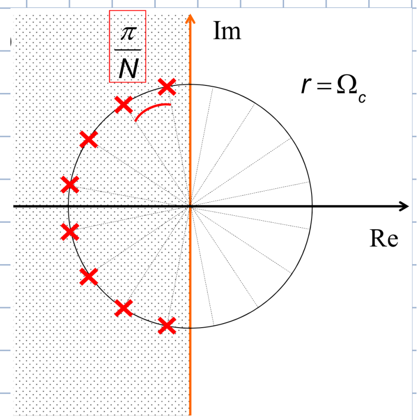
ALL POLES ON CIRCLE OF RADIUS Ω_c

TO FIND LOCATIONS NEED TO FIND PHASE:

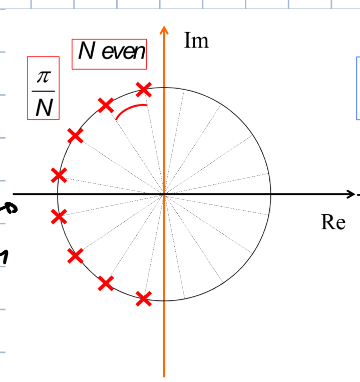
$$\frac{\pi}{2N}(2k-1+N) = \underbrace{\frac{\pi}{2N}2k}_{\text{varies}} \underbrace{- \frac{\pi}{2N} + \frac{\pi}{2}}_{\text{basically offset}}$$ → ONLY VARIABLE IS K

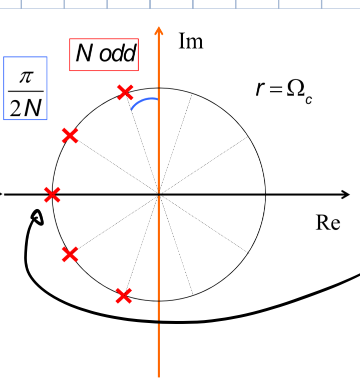
$$\frac{\pi}{2N}2k = \pi\frac{k}{N}\Big|_{k=1} = \frac{\pi}{N}$$ → NEED TO DIVIDE PERIOD OF 2π INTO $2N$ POLES SEPARATED BY A $\frac{\pi}{N}$ PHASE

REMEMBER THAT THIS IS $|H(s)|^2$ AND WE WANT $|H(s)|$.
$|H(s)|^2$ CAN BE REPRESENTED AS $H(s)H(-s)$, WHICH MEANS WE CAN CHOOSE TO TAKE JUST THE POLES ON $Re\{s\} < \emptyset$ RESULTING IN A STABLE FILTER



$$H(s) = \frac{\Omega_c^N}{\prod_{k=1}^{N}(s - s_k)}$$





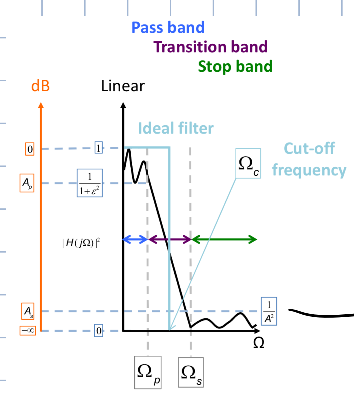
IF EVEN, NEVER POLE IN -1

IF ODD, ALWAYS POLE IN -1

WE NOW NEED A WAY TO GO FROM THE SPECS TO THE ACTUAL DESIGN OF THE FILTER.

$$|H(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

$$\frac{1}{1 + \left(\frac{\Omega_p}{\Omega_c}\right)^{2N}} \geq \frac{1}{1+\varepsilon^2} = A_p$$



$$\frac{1}{1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N}} \leq \frac{1}{A^2} = A_s$$

$$\cancel{1}+\varepsilon^2 \geq \cancel{1}+\left(\frac{\Omega_p}{\Omega_c}\right)^{2N} \qquad\qquad 1+\left(\frac{\Omega_s}{\Omega_c}\right)^{2N} \geq A^2$$

$$\Big\downarrow \tfrac{1}{x} \qquad\qquad\qquad\qquad \Big\downarrow$$

$$\frac{1}{\left(\frac{\Omega_p}{\Omega_c}\right)^{2N}} \geq \frac{1}{\varepsilon^2} \qquad\qquad \left(\frac{\Omega_s}{\Omega_c}\right)^{2N} \geq A^2-1$$

$$\Big\downarrow 10\log_{10} \qquad\qquad\qquad\qquad \Big\downarrow 10\log_{10}$$

$$\underbrace{-10\log_{10}\left[\left(\frac{\Omega_p}{\Omega_c}\right)^{2N}\right] \geq -10\log_{10}(\varepsilon^2) \qquad\qquad 10\log_{10}\left[\left(\frac{\Omega_s}{\Omega_c}\right)^{2N}\right] \geq 10\log_{10}(A^2-1)}_{+}$$

$$10\log_{10}\left[\left(\frac{\Omega_s}{\Omega_c}\right)^{2N}\right]-10\log_{10}\left[\left(\frac{\Omega_p}{\Omega_c}\right)^{2N}\right] \geq 10\log_{10}(A^2-1)-10\log_{10}(\varepsilon^2)$$

$$2N\left[10\log_{10}\left(\frac{\Omega_s}{\Omega_c}\right)-10\log_{10}\left(\frac{\Omega_p}{\Omega_c}\right)\right] \geq 10\log_{10}\left(\frac{A^2-1}{\varepsilon^2}\right)$$

$$2N\cdot\cancel{10}\log_{10}\left(\frac{\Omega_s}{\cancel{\Omega_c}}\,\frac{\cancel{\Omega_c}}{\Omega_p}\right) \geq \cancel{10}\log_{10}\left(\frac{A^2-1}{\varepsilon^2}\right)$$

DESIGN EQUATION

$$\boxed{N \geq \frac{1}{2}\,\frac{\log_{10}\left(\frac{A^2-1}{\varepsilon^2}\right)}{\log_{10}\left(\frac{\Omega_s}{\Omega_p}\right)}}$$

TELLS US MINIMUM ORDER OF FILTER TO SATISFY REQUIREMENTS

NOTE THAT IF Ω_s AND Ω_p ARE TOO CLOSE, THE DENOMINATOR GOES TO ZERO AND $N \to \infty$.

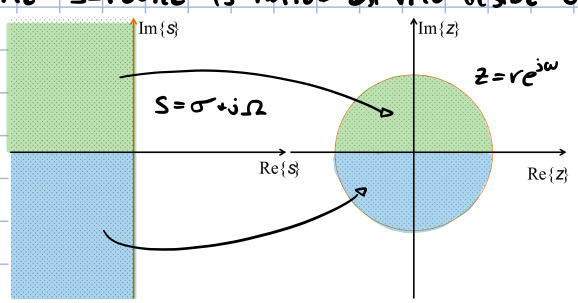
IN CONCLUSION, SUPPOSE YOU WANT TO DESIGN A BUTTERWORTH LPF WITH SPECIFIED $\Omega_p, \Omega_s, A_p, A_s$, THE DESIGN PROCESS CONSISTS OF DETERMINING N AND Ω_c. AFTER THIS, THE EQUATION IS KNOWN.

BILINEAR TRANSFORM

THERE ARE SEVERAL METHODS TO CONVERT FROM CONTINUOUS TIME FILTERS (ANALOG PROTOTYPES) TO DISCRETE ONES. WE'LL ONLY DISCUSS THE BILINEAR TRANSFORM.

IT IS A MAP FROM AND TO THE S AND Z PLANES.

ANY USEFUL MAPPING SHOULD SATISFY 3 DESIRABLE CONDITIONS:
1) A RATIONAL $H(s)$ SHOULD MAP TO A RATIONAL $H(z)$
2) THE IMAGINARY AXIS OF THE S-PLANE IS MAPPED ON THE UNIT CIRCLE IN Z
3) THE LEFT HALF OF THE S-PLANE IS MAPPED ON THE INSIDE OF THE UNIT CIRCLE IN Z



$$\boxed{S = \frac{2}{T}\frac{z-1}{z+1}} = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$$, T = SAMPLING PERIOD

$$\boxed{z = \frac{1+s\frac{T}{2}}{1-s\frac{T}{2}}}$$

Ω - ω LINK

WE KNOW HOW TO LINK $\omega[\text{rad}/s]$ TO $f[\text{Hz}]$. LET'S EXPLORE NOW HOW TO LINK Ω TO ω.

$$\sigma := \varnothing \implies s = j\Omega$$

$$s = \frac{2}{T}\frac{z-1}{z+1} \longrightarrow s(z+1)\frac{T}{2} = z-1 \longrightarrow s\frac{T}{2}z + s\frac{T}{2} = z-1 \longrightarrow s\frac{T}{2}+1 = z - s\frac{T}{2}z \longrightarrow 1+s\frac{T}{2} = z\left(1-s\frac{T}{2}\right) \longrightarrow z = \frac{1+s\frac{T}{2}}{1-s\frac{T}{2}}$$

$$z = \frac{1+s\frac{T}{2}}{1-s\frac{T}{2}} = \frac{1+j\Omega\frac{T}{2}}{1-j\Omega\frac{T}{2}}$$

$z = e^{j\omega}$ IMAG. AXIS IN S = UNIT CIRCLE IN Z

$$s = j\Omega = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$$

$$j\Omega = \frac{2}{T}\frac{1-e^{-j\omega}}{1+e^{-j\omega}} = \frac{2}{T}\frac{\cancel{e^{-\frac{1}{2}j\omega}}}{\cancel{e^{-\frac{1}{2}j\omega}}}\frac{e^{\frac{1}{2}j\omega}-e^{-\frac{1}{2}j\omega}}{e^{\frac{1}{2}j\omega}+e^{-\frac{1}{2}j\omega}} = \frac{2}{T}\frac{\cancel{2}j\sin(\omega/2)}{\cancel{2}\cos(\omega/2)} = \frac{2}{T}j\tan\left(\frac{\omega}{2}\right)$$

$$\cancel{j}\Omega = \frac{2}{T}\cancel{j}\tan\left(\frac{\omega}{2}\right) \longrightarrow \boxed{\Omega = \frac{2}{T}\tan\left(\frac{\omega}{2}\right) \longleftrightarrow \omega = 2\tan^{-1}\left(\frac{\Omega T}{2}\right)}$$

EX

$$0.89125 \le |H(e^{j\omega})| \le 1 \quad , \quad 0 \le |\omega| \le 0{,}2\pi$$

$$|H(e^{j\omega})| \le 0.17783 \quad , \quad 0{,}3\pi \le |\omega| \le \pi$$

1) OBTAIN EQUIVALENT MARGINS IN S DOMAIN

$$0.89125 \le |H(e^{j\omega})| \le 1 \quad , \quad 0 \le |\Omega| \le \frac{2}{T}\tan\left(\frac{0{,}2\pi}{2}\right) \rightarrow \Omega_p = \frac{2}{T}\tan\left(\frac{0{,}2\pi}{2}\right)$$

$$|H(e^{j\omega})| \le 0.17783 \quad , \quad \frac{2}{T}\tan\left(\frac{0{,}3\pi}{2}\right) \le |\Omega| \le \infty \rightarrow \Omega_s = \frac{2}{T}\tan\left(\frac{0{,}3\pi}{2}\right)$$

2) FIND ORDER N AND CUTOFF STARTING FROM FORMULA AND FILLING IN NUMBERS

$$|H(j\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

$$\frac{1}{1+\left(\frac{\Omega_p}{\Omega_c}\right)^{2N}} = 0{,}89125^2 \longrightarrow \frac{1}{1+\left(\frac{\frac{2}{T}\tan\left(\frac{0{,}2\pi}{2}\right)}{\Omega_c}\right)^{2N}} = 0{,}89125^2$$

$$\frac{1}{1+\left(\frac{\Omega_s}{\Omega_c}\right)^{2N}} = 0{,}17783^{2} \longrightarrow \frac{1}{1+\left(\frac{\frac{2}{T}\tan\left(\frac{0{,}3\pi}{2}\right)}{\Omega_c}\right)^{2N}} = 0{,}17783^{2}$$

SUPP. $T=1$

$$0.89125^{-2} = 1+\left(\frac{2\tan(0.1\pi)}{\Omega_c}\right)^{2N}$$

$$0.17783^{-2} = 1+\left(\frac{2\tan(0.15\pi)}{\Omega_c}\right)^{2N}$$

WE CAN FIND N VIA THE DESIGN EQUATION

$$N \geqslant \frac{1}{2}\,\frac{\log_{10}\left(\frac{A^2-1}{\varepsilon^2}\right)}{\log_{10}\left(\frac{\Omega_s}{\Omega_p}\right)}$$

$$\frac{1}{A^2} = 0.17783^{2} \longrightarrow A^2 = 0.17783^{-2}$$

$$\frac{1}{1+\varepsilon^2} = 0.89125^{2} \longrightarrow \varepsilon^2 = 0.89125^{-2}-1$$

$$\frac{\log_{10}\left(\frac{0.17783^{-2}-1}{0.89125^{-2}-1}\right)}{\log_{10}\left(\frac{2\tan(0.15\pi)}{2\tan(0.1\pi)}\right)} \approx 5.3 \longrightarrow N \geqslant 6$$

NOW WE CAN COMPUTE Ω_c FROM EITHER OF THE TWO EQUATIONS

$$0.17783^{-2} = 1+\left(\frac{2\tan(0.15\pi)}{\Omega_c}\right)^{2N} = 1+\left(\frac{2\tan(0.15\pi)}{\Omega_c}\right)^{2\cdot 6} \longrightarrow \Omega_c = \frac{2\tan(0.15\pi)}{\left(0.17783^{-2}-1\right)^{\frac{1}{12}}} \approx 0.766$$

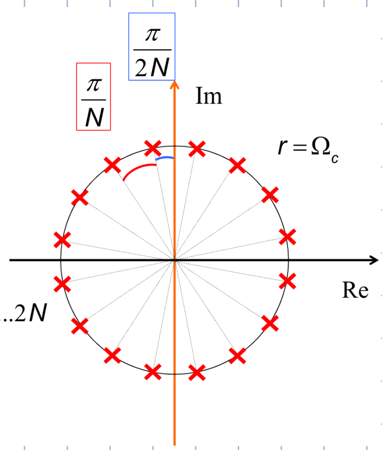
NOW THAT WE KNOW THE RADIUS OF THE CIRCLE, WE CAN FIND THE POLES

$$|H(s)|^2 = \frac{1}{1+\left(\frac{s}{j\Omega_c}\right)^{2N}}$$

poles in

$$\left(\frac{s}{j\Omega_c}\right)^{2N} = -1 = e^{j(2k\pi-\pi)} \qquad k=1,\dots,2N$$

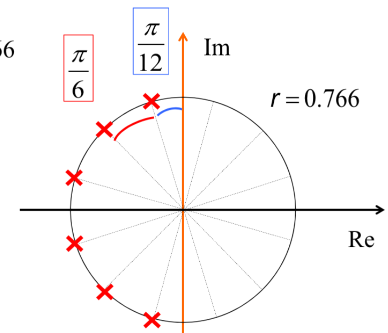
$$s = e^{j\frac{(-\pi+2k\pi)}{2N}}\, j\Omega_c = e^{j\frac{\pi}{2N}(2k-1+N)}\,\Omega_c$$



$$s = e^{j\frac{\pi}{2N}(2k-1+N)}\,\Omega_c = e^{j\frac{\pi}{12}(2k+5)}\,0.766$$

$k \in [1,12]$

Remember that we keep only
The poles with Re{s} negative



NOW WRITE $H(S)$

$$H(s) = \frac{a_0}{(s-s_1)(s-s_2)(s-s_3)(s-s_4)(s-s_5)(s-s_6)} = \frac{a_0}{(s-s_1)(s-s_2)(s-s_3)(s-s_1^*)(s-s_2^*)(s-s_3^*)}$$

POLES ARE COMPLEX CONJUGATE

$$= \frac{a_0}{\left(s^2 - 2\mathrm{Re}\{s_1\}s + |s_1|^2\right)\left(s^2 - 2\mathrm{Re}\{s_2\}s + |s_2|^2\right)\left(s^2 - 2\mathrm{Re}\{s_3\}s + |s_3|^2\right)}$$

$$= \frac{\Omega_c^N}{\left(s^2 - 2\Omega_c \cos\left[\frac{\pi 7}{12}\right]s + |\Omega_c|^2\right)\left(s^2 - 2\Omega_c \cos\left[\frac{\pi 9}{12}\right]s + |\Omega_c|^2\right)\left(s^2 - 2\Omega_c \cos\left[\frac{\pi 11}{12}\right]s + |\Omega_c|^2\right)}$$

$$= \frac{\Omega_c^N}{\left(s^2 - 2\Omega_c \cos\left[\frac{\pi 7}{12}\right]s + |\Omega_c|^2\right)\left(s^2 - 2\Omega_c \cos\left[\frac{\pi 9}{12}\right]s + |\Omega_c|^2\right)\left(s^2 - 2\Omega_c \cos\left[\frac{\pi 11}{12}\right]s + |\Omega_c|^2\right)} \xrightarrow{\text{VIA BILINEAR}} H(z)$$

DIFFERENT FILTER TYPES

WE'VE ONLY SEEN LPF, BUT WE CAN CONVERT TO BPF, HPF, ... USING FREQUENCY TRANSFORMS. WHEN USING THE BILINEAR TRANSFORM, THESE FREQUENCY TRANSFORMATIONS MAY BE PERFORMED BEFORE OR AFTER.

Filter type	Transformation	Design parameters
Lowpass	$z^{-1} \to \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$	ω_c = cutoff frequency $\alpha = \frac{\sin[(\theta_c - \omega_c)/2]}{\sin[(\theta_c + \omega_c)/2]}$
Highpass	$z^{-1} \to -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$	ω_c = cutoff frequency $\alpha = -\frac{\cos[(\theta_c + \omega_c)/2]}{\cos[(\theta_c - \omega_c)/2]}$
Bandpass	$z^{-1} \to -\frac{z^{-2} - \alpha_1 z^{-1} + \alpha_2}{\alpha_2 z^{-2} - \alpha_1 z^{-1} + 1}$	ω_1 = lower cutoff frequency ω_2 = upper cutoff frequency $\alpha = \frac{\cos[(\omega_2 + \omega_1)/2]}{\cos[(\omega_2 - \omega_1)/2]}$ $K = \cot[(\omega_2 - \omega_1)/2]\tan(\theta_c/2)$ $\alpha_1 = 2\alpha K/(K+1)$ $\alpha_2 = (K-1)/(K+1)$
Bandstop	$z^{-1} \to \frac{z^{-2} - \alpha_1 z^{-1} + \alpha_2}{\alpha_2 z^{-2} - \alpha_1 z^{-1} + 1}$	ω_1 = lower cutoff frequency ω_2 = upper cutoff frequency $\alpha = \frac{\cos[(\omega_2 + \omega_1)/2]}{\cos[(\omega_2 - \omega_1)/2]}$ $K = \tan[(\omega_2 - \omega_1)/2]\tan(\theta_c/2)$ $\alpha_1 = 2\alpha/(K+1)$ $\alpha_2 = (1-K)/(1+K)$

θ_c = LPF CUTOFF

IIR VS. FIR

IN IIR WE HAVE NO CONTROL ON PHASE, ONLY MAGNITUDE. WITH FIR WE CAN HAVE LINEAR PHASE.

FIRs OFTEN REQUIRE MUCH HIGHER ORDERS TO ACHIEVE SAME PERFORMANCE OF IIR.
IF PHASE IS NOT A CONCERN, IIRs ARE PREFERRABLE